

Setup Test Directory

```
Ubuntu x + -
GNU nano 7.2 /tmp/crontab.L6nnC6/crontab
2 * * * rsync -av ~/backup-demo/source/ ~/backup-demo/rsync-dest/
# Edit this file to introduce tasks to be run by cron.
#
```

Full Backup using tar -

- c create
- v verbose
- z gzip
- f filename

```
priyansh@IDEAPAD-18UT:~$ tar -cvzf ~/backup-demo/backups/full_backup_$(date +%F).tar.gz ~/backup-demo/source/
priyansh@IDEAPAD-18UT:~$ tar -cvzf ~/backup-demo/backups/full_backup_$(date +%F).tar.gz ~/backup-demo/source/
tar: Removing leading '/' from member names
/home/priyansh/backup-demo/source/
/home/priyansh/backup-demo/source/file2.txt
/home/priyansh/backup-demo/source/file1.txt
```

Verify the backup

```
priyansh@IDEAPAD-18UT:~$ tar -tvzf ~/backup-demo/backups/full_backup_$(date +%F).tar.gz
drwxr-xr-x priyansh/priyansh 0 2026-04-23 23:13 home/priyansh/backup-demo/source/
-rw-r--r-- priyansh/priyansh 7 2026-04-23 23:13 home/priyansh/backup-demo/source/file2.txt
-rw-r--r-- priyansh/priyansh 7 2026-04-23 23:13 home/priyansh/backup-demo/source/file1.txt
```

Stimulate an incremental change and then incremental backup (only newer files)

```
priyansh@IDEAPAD-18UT:~$ echo "New File 3" > ~/backup-demo/source/file3.txt
priyansh@IDEAPAD-18UT:~$ # Find files modified in last 1 day and back them up
find ~/backup-demo/source/ -newer ~/backup-demo/backups/full_backup_$(date +%F).tar.gz \
| tar -cvzf ~/backup-demo/backups/incremental_$(date +%F_%T).tar.gz -T -
tar: Removing leading '/' from member names
/home/priyansh/backup-demo/source/
/home/priyansh/backup-demo/source/file2.txt
tar: Removing leading '/' from hard link targets
/home/priyansh/backup-demo/source/file3.txt
/home/priyansh/backup-demo/source/file1.txt
/home/priyansh/backup-demo/source/file3.txt
priyansh@IDEAPAD-18UT:~$
```